CV

CostVision

Manufacturing Should-Cost Intelligence Platform

SHOULD-COST ANALYSIS REPORT

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Bottom-Up Manufacturing Cost Model · Fully Traceable Rate Data

Cast Suspension/Drivetrain Lever Arm (Yoke-type component)

Commodity: CASTING · Currency: CNY · FX Rate: 9.0498 to GBP · Operations: 2 · Region: CN

Total Should-Cost	Material	Process	Labour	Margin
¥301.97	**12.2%**	**48.5%**	**15.0%**	**7.4%**

Model Confidence: **Low** · 2 traced operations · 7 data points auditable

Engine Warnings (1)

rawMaterial.materialId: Material rate confidence: Low

Geometry Provenance — MEASURED (OCCT B-rep kernel)

Volume, weight and every feature were measured from the CAD solid by the Open CASCADE kernel. Measured volume 429.4 cm³; mass for the selected material family 3.092 kg.
Material cost and geometry-derived machining are grounded in the actual solid — not an estimate.

Key Assumptions

Annual volume: 200,000 · Region: CN · Commodity: casting
Alloy / material: EN-GJL-250 · Net weight: 3.165 kg (measured)
Material utilisation: 65% · Overhead: 12% of factory base · Margin: 8% of subtotal · Operations: 2

§1 — 8-Bucket Cost Breakdown

Cost Bucket	Amount (CNY)	% of Total	Cost Mix Bar
1. Raw Material	¥36.76	12.2%	
2. Process (Machine)	¥146.43	48.5%	
3. Direct Labour	¥45.45	15.0%	
4. Tooling (amortised)	¥16.97	5.6%	
5. Packaging	¥1.63	0.5%	

Cost Bucket	Amount (CNY)	% of Total	Cost Mix Bar
6. Logistics	¥2.90	1.0%	
Factory Cost	**¥250.13**	**82.8%**	
7. Overhead (SG&A)	¥29.47	9.8%	
Subtotal	**¥279.61**	**92.6%**	
8. Supplier Margin	¥22.37	7.4%	
TOTAL SHOULD-COST	**¥301.97**	**100.0%**	

§2 — Commercial Parameters

Overhead Rate	**12.0%**	Supplier Margin	**8.0%**
Packaging / part	**¥1.63**	Logistics / part	**¥2.90**
Total Tooling Cost	**¥3393665.16**	Amortisation Volume	**200,000 parts**

Geometric DFM / DFA - measured from the CAD model

3 issue(s) across 54 instance(s), from 92 measured feature(s) and 4 rule(s). Every issue names the B-rep faces that produced it and the published source of its threshold. No finding on this part has a modelled cost path; each says why.

[MAJOR] Wall below minimum draft for the casting route (36 instances)

Worst: Face 11 draws at 0.00° against a 0.5° minimum for this route.
measured draftDeg 0-0° threshold < 0.5°
Faces: 6, 7, 8, 9, 11, 12, 13, 14, 19, 22, 75, 76, 87, 88, 89, 90, 91, 93, 95, 96, 97, 98, 153, 161, 175, 182, 194, 195, 216, 217 ... (36 total)
Fix: Add draft to at least 0.5° per side, or accept the ejection drag and the die wear it causes.
Source: ASM Handbook Vol. 15, Casting — pattern and die draft
Route not stated, so the most permissive published minimum (0.5°) is applied — this under-reports rather than over-reports when the process is unknown.

[MAJOR] Undercut face — needs a slide or a core (12 instances)

Worst: Face 129 sits at 158.2° to the draw (past 90°, so its normal opposes withdrawal) and cannot release on the main parting.
measured angleToDrawDeg 112-158.2° threshold > 90°
Faces: 27, 28, 31, 38, 40, 42, 129, 130, 156, 160, 178, 179
Fix: Re-orient the parting line, or price a slide/core for this feature and carry the die cost and cycle penalty explicitly.
Source: NADCA Product Specification Standards for Die Castings - Undercuts and moving die components
Any face whose normal opposes the draw cannot release on the main parting; it requires a slide, a loose piece or a core, each of which adds die cost and cycle time.

[MAJOR] Abrupt section change (6 instances)

Worst: Section steps 3.62:1 at face 153 (27.1 mm against an adjacent 7.5 mm).
measured sectionRatio 2.925-3.618:1 threshold > 2:1
Faces: 87, 88, 89, 90, 153, 175
Fix: Blend the transition with a taper or a generous fillet so the section changes gradually; expect shrinkage porosity at the junction otherwise.
Source: ASM Handbook Vol. 15, Casting — design for solidification
A thick-to-thin ratio above ~2:1 across a junction concentrates solidification shrinkage and residual stress. Widely published as a 2:1 guideline; blended transitions are the standard remedy.

DFA - handling & insertion (Boothroyd method, geometric half)

Handling 1.13s + insertion 2.9s = 4.03s, 1.34x the 3s ideal part.
+1.4s Fasteners approach from more than one direction — the part must be re-oriented or the fixture indexed during assembly (3 distinct hole axes across 14 holes)
Theoretical minimum part count, and therefore the Boothroyd design-efficiency index, requires the full assembly — this analysis covers one part, so no efficiency ratio is given.
Nesting and tangling risk is not assessed: it depends on how parts are presented in bulk, which the geometry alone does not determine.
Rotational symmetry is approximated from the bounding box, not from an inertia-tensor shape comparison.

What was NOT checked

No material family was confirmed, so any check whose threshold depends on the alloy or resin did not run.
Tolerance and datum callouts were not checked — they live on the drawing, not in the solid.
Parting-line position was assumed to lie perpendicular to the draw direction; a different parting choice changes which faces are undercuts.
Machining stock allowance on cast surfaces was not assessed — it is a process decision, not a measurable feature.
Gating and feeding layout was not evaluated; hot spots are reported, but where to place risers is a foundry decision.

§3 — Material Detail

Parameter	Value	Unit	Notes
Material ID	**mat-gjl250**	ID	
Grade / Specification	**EN-GJL-250**		UK iron foundry, Jun 2026. Index-anchored 2026-07 (see CASTING PRICING BASIS). | Regional adj. ×0.880 (millSteel)
Region	**China**		
Net Finished Weight	**3.1649 kg**	kg	Weight in finished part
Gross Weight (stock)	**4.8692 kg**	kg	Net ÷ utilisation ratio
Scrap / Runner Weight	**1.7042 kg**	kg	Gross - Net
Material Utilisation	**65.0%**	%	Benchmark: casting 65–85 % (runners + risers)
Material Price	**¥5.10**	CNY /kg	UK iron foundry, Jun 2026. Index-anchored 2026-07 (see CASTING PRICING BASIS). | Regional adj. ×0.880 (millSteel)
Scrap Recovery Price	**¥0.96**	CNY /kg	
Gross Material Cost	**¥24.82**	CNY	Gross × price/kg
Scrap Credit	**-¥1.63**	CNY	Scrap × recovery price
Consumables (core / wax / shell)	**¥13.57**	CNY	Per-part recurring
NET RAW MATERIAL COST	**¥36.76**	CNY	Gross - scrap credit + consumables

Data Confidence	**Low**		
Effective Date	**2026-07**		

§4 — Operations Detail

4A Machine Operations

Operation	Machine Class	Rate / hr	Cycle (min)	Parts / Cycle	OEE %	Machine Cost
Sand Casting — Moulding	Sand Casting Moulding Line	¥136.44	12.15	1	80.0%	**¥34.53**
CNC Machining — 29 features (2×Ø8.0×18, 2×Ø8.0×23, 2×Ø10.5×13, 2×Ø14.0×10, 2×Ø22.0×40, 1×Ø24.0×16, 1×Ø43.0×8, 1×Ø50.0×13, 2×Ø50.0×32, 2× face	CNC VMC 3-axis	¥254.30	22.44	1	85.0%	**¥111.90**

Operation	Machine Class	Rate / hr	Cycle (min)	Parts / Cycle	OEE %	Machine Cost
2915mm², 2× face 2551mm², 2× face 1986mm², 2× face 1423mm², 1× face 1357mm², 1× face 868mm², 2× face 698mm², 2× face 695mm²) [geometry-measured]						
TOTAL						**¥146.43**

4B Labour Detail

Operation	Labour Grade	Rate / hr	Man ning	Lab. min	Eff %	Labour Cost	Op Total
Sand Casting — Moulding	Skilled Machinist	¥71.49	1	12.15	92.0 %	¥15.73	**¥50.27**
CNC Machining — 29 features (2×Ø8.0×18, 2×Ø8.0×23, 2×Ø10.5×13, 2×Ø14.0×10, 2×Ø22.0×40, 1×Ø24.0×16, 1×Ø43.0×8, 1×Ø50.0×13, 2×Ø50.0×32, 2× face 2915mm², 2× face 2551mm², 2× face 1986mm², 2× face 1423mm², 1× face 1357mm², 1× face 868mm², 2× face 698mm², 2× face 695mm²) [geometry-measured]	Skilled Machinist	¥71.49	1	22.44	90.0 %	¥29.71	**¥141.61**
TOTAL						**¥45.45**	**¥191.88**

4C Machined Features — Geometry Audit

Feature	Ø×depth / area	Qty	Operation	Min	Cost	Costed ?
Hole / bore	Ø8.0 × 18 blind	2	Drilling (blind)	1.20	¥5.10	Yes
Hole / bore	Ø8.0 × 23 blind	2	Drilling (blind)	1.44	¥6.09	Yes
Hole / bore	Ø10.5 × 13 blind	2	Drilling (blind)	1.03	¥4.36	Yes
Hole / bore	Ø14.0 × 10 blind	2	Drill + bore (blind)	1.80	¥7.63	Yes
Hole / bore	Ø22.0 × 40 blind	2	Drill + bore (blind)	4.17	¥17.67	Yes
Hole / bore	Ø24.0 × 16 blind	1	Drill + bore (blind)	1.14	¥4.83	Yes
Hole / bore	Ø43.0 × 8 blind	1	Drill + bore (blind)	0.37	¥1.58	Yes
Hole / bore	Ø50.0 × 13 blind	1	Drill + bore (blind)	1.25	¥5.28	Yes
Hole / bore	Ø50.0 × 32 blind	2	Drill + bore (blind)	4.40	¥18.65	Yes
Face	2915 mm²	2	Face milling (facing)	1.13	¥4.79	Yes
Face	2551 mm²	2	Face milling (facing)	1.04	¥4.40	Yes
Face	1986 mm²	2	Face milling (facing)	0.90	¥3.80	Yes
Face	1423 mm²	2	Face milling (facing)	0.76	¥3.20	Yes
Face	1357 mm²	1	Face milling (facing)	0.37	¥1.57	Yes

Feature	Ø×depth / area	Qty	Operation	Min	Cost	Costed ?
Face	868 mm²	1	Face milling (facing)	0.31	¥1.31	Yes
Face	698 mm²	2	Face milling (facing)	0.57	¥2.44	Yes
Face	695 mm²	2	Face milling (facing)	0.57	¥2.43	Yes
Pocket	2915 mm² × 22	2	Pocket milling (rough + finish)	22.97	—	No
Pocket	1986 mm² × 26	2	Pocket milling (rough + finish)	18.72	—	No
TOTAL costed		**29**		**22.44**	**¥95.11**	**Yes**

4 detected feature(s) NOT costed

4 machinable feature instance(s) (41.7 min) were detected in the geometry but are NOT included in this cost — on a near-net part they may be as-cast. If they are machined, tick them in the Machined-Features panel and re-cost; this could add up to ¥176.71.

§5 — Machine Rate Buildup

CNC VMC 3-axis [mach-vmc3] · Confidence: Low · UK Tier-2 3-axis VMC (HAAS VF/VM class). Target £55/hr. Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @80% util	Notes
Depreciation	¥273755.66	¥85.55	4,000 hr/yr available
Maintenance	¥149321.27	¥46.66	
Energy	¥77119.81	¥24.10	
Floor Space	¥59728.51	¥18.67	
Indirect Support	¥149321.27	¥46.66	
Finance Cost	¥104524.89	¥32.66	
TOTAL — CNC VMC 3-axis	¥813771.39	¥254.30	Effective hours: 3200/yr

Sand Casting Moulding Line [sand-cast-line] · Confidence: Low · UK foundry benchmark, Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @75% util	Notes
Depreciation	¥124434.39	¥47.40	3,500 hr/yr available
Maintenance	¥59728.51	¥22.75	
Energy	¥68856.97	¥26.23	
Floor Space	¥49773.76	¥18.96	
Indirect Support	¥39819.00	¥15.17	
Finance Cost	¥15554.30	¥5.93	
TOTAL — Sand Casting Moulding Line	¥358166.93	¥136.44	Effective hours: 2625/yr

How to read the machine rate

Rate/hr = (annual depreciation + maintenance + energy + floor + indirect + finance) ÷ effective hours, where effective hours = available hours × utilisation.

Effective hours encode the shift pattern: a lower hours-base raises £/hr and a higher one lowers it. A challenge from a supplier will target this divisor and whether depreciation is machine-only or full-cell (machine + dosing furnace + trim + robots/extraction) — state your basis when defending the number.

§6 — Rate Traceability

Field	Value	Unit	Source / Reference	Rate ID	Conf.
material.pricePerKg	0.5632	£/kg	UK iron foundry, Jun 2026. Index-anchored 2026-07 (see CASTING PRICING BASIS). | Regional adj.	mat-gjl250	Low

Field	Value	Unit	Source / Reference	Rate ID	Conf.
			×0.880 (millSteel)		
material.scrapRecoveryPricePerKg	0.1056	£/kg	UK iron foundry, Jun 2026. Index-anchored 2026-07 (see CASTING PRICING BASIS). | Regional adj. ×0.880 (millSteel)	mat-gjl250	Low
rawMaterial.consumablesCostPerPart	1.5000	£	Per-part consumable (core/wax/shell)	mat-gjl250	Medium
Sand Casting — Moulding.machineRatePerHr	15.0771	£/hr	UK foundry benchmark, Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	sand-cast-line	Low
Sand Casting — Moulding.labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low
CNC Machining — 29 features (2×Ø8.0×18, 2×Ø8.0×23, 2×Ø10.5×13, 2×Ø14.0×10, 2×Ø22.0×40, 1×Ø24.0×16, 1×Ø43.0×8, 1×Ø50.0×13, 2×Ø50.0×32, 2× face 2915mm², 2× face 2551mm², 2× face 1986mm², 2× face 1423mm², 1× face 1357mm², 1× face 868mm², 2× face 698mm², 2× face 695mm²) [geometry-measured].machineRatePerHr	28.1005	£/hr	UK Tier-2 3-axis VMC (HAAS VF/VM class). Target £55/hr. Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	mach-vmc3	Low
CNC Machining — 29 features (2×Ø8.0×18, 2×Ø8.0×23, 2×Ø10.5×13, 2×Ø14.0×10, 2×Ø22.0×40, 1×Ø24.0×16, 1×Ø43.0×8, 1×Ø50.0×13, 2×Ø50.0×32, 2× face 2915mm², 2× face 2551mm², 2× face 1986mm², 2× face 1423mm², 1× face 1357mm², 1× face 868mm², 2× face 698mm², 2× face 695mm²) [geometry-measured].labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low

§7 — Should-Cost with Uncertainty

Monte-Carlo · Low confidence

Estimate (± band)	P10 · optimistic	P50 · median	P90 · conservative	Band	CV
¥301.97 ± 18.2%	¥249.32	¥299.00	¥359.37	wide	14.3%

Range reflects how well each cost driver is known (High / Medium / Low confidence). Tighten it by attaching CAD, a BOM, or logging an actual quote.

Why confidence is Low

Main band drivers: 6 rate(s) at Low confidence; tooling amortisation carries the widest per-bucket spread; some detected features not costed (see §4C).

What's excluded from this unit cost

One-time NRE — tooling / die, fixtures and CNC programming — is amortised separately, not in this unit cost.
Import duty and international freight (regional table is Ex-Works).
Formal embodied-carbon reporting (figures are indicative cradle-to-gate — replace with supplier EPDs).
Heat-treat (T6/T7), HIP and 100% NDT/X-ray are NOT in this cost — add them for structural / safety-critical parts.
Trim/degate, shot-blast/deflash and impregnation are only included where shown as operations.

§8 — Sensitivity Analysis

± 10% driver swing · ranked by ¥ impact

Cost Driver	Base	-10% cost	+10% cost	Range ¥
CNC Machining — 29 features (2×Ø8.0×18, 2×Ø8.0×23, 2×Ø10.5×13, 2×Ø14.0×10, 2×Ø22.0×40, 1×Ø24.0×16, 1×Ø43.0×8, 1×Ø50.0×13, 2×Ø50.0×32, 2× face 2915mm², 2× face 2551mm², 2× face 1986mm², 2× face 1423mm², 1× face 1357mm², 1× face 868mm², 2× face 698mm², 2× face 695mm²) [geometry-measured]: Cycle Time	0.37hr	-5.7%	+5.7%	**¥34.26**
CNC Machining — 29 features (2×Ø8.0×18, 2×Ø8.0×23, 2×Ø10.5×13, 2×Ø14.0×10, 2×Ø22.0×40, 1×Ø24.0×16, 1×Ø43.0×8, 1×Ø50.0×13, 2×Ø50.0×32, 2× face 2915mm², 2× face 2551mm², 2× face 1986mm², 2× face 1423mm², 1× face 1357mm², 1× face 868mm², 2× face 698mm², 2× face 695mm²) [geometry-measured]: Machine Rate	¥254.30/hr	-4.5%	+4.5%	**¥27.07**
Sand Casting — Moulding: Cycle Time	0.2hr	-2%	+2%	**¥12.16**
Sand Casting — Moulding: Labour Rate	¥71.49/hr	-1.8%	+1.8%	**¥10.99**
CNC Machining — 29 features (2×Ø8.0×18, 2×Ø8.0×23, 2×Ø10.5×13, 2×Ø14.0×10, 2×Ø22.0×40, 1×Ø24.0×16, 1×Ø43.0×8, 1×Ø50.0×13, 2×Ø50.0×32, 2× face 2915mm², 2× face 2551mm², 2× face 1986mm², 2× face 1423mm², 1× face 1357mm², 1× face 868mm², 2× face 698mm², 2× face 695mm²) [geometry-measured]: Labour Rate	¥71.49/hr	-1.8%	+1.8%	**¥10.99**
Sand Casting — Moulding: Machine Rate	¥136.44/hr	-1.4%	+1.4%	**¥8.35**
Overhead %	12%	-1.1%	+1.1%	**¥6.37**
Material: EN-GJL-250	¥5.10/kg	-1%	+1%	**¥6.00**

§9 — Regional Cost Comparison

Ex-Works · per-region should-cost · vs China

Region	Material	Process	Labour	Tooling	Overhead	Ex-Works	Logistics	Total	vs China
United Kingdom (GBP)	¥41.78	¥266.24	¥163.60	¥30.85	¥39.30	**¥541.77**	¥2.00	**¥568.46**	**+88%**
Germany (EUR)	¥43.03	¥279.55	¥264.41	¥32.39	¥53.29	**¥672.67**	¥2.30	**¥705.16**	**+134%**
France (EUR)	¥42.61	¥244.94	¥190.04	¥28.38	¥41.55	**¥547.53**	¥2.30	**¥574.89**	**+90%**
Spain (EUR)	¥41.78	¥234.29	¥119.81	¥27.15	¥31.43	**¥454.46**	¥2.30	**¥477.93**	**+58%**
Poland (PLN)	¥40.52	¥191.69	¥74.36	¥22.21	¥21.86	**¥350.65**	¥2.40	**¥369.69**	**+22%**
Turkey (TRY)	¥37.60	¥159.74	¥41.31	¥18.51	¥15.69	**¥272.85**	¥2.60	**¥288.67**	**-4%**
China (CNY)	¥36.76	¥146.43	¥45.45	¥16.97	¥29.47	**¥275.08**	¥2.90	**¥301.97**	**Base**
India (INR)	¥37.60	¥138.44	¥28.92	¥16.04	¥12.44	**¥233.45**	¥3.00	**¥247.71**	**-18%**
Mexico (MXN)	¥39.69	¥159.74	¥47.92	¥18.51	¥16.22	**¥282.08**	¥2.70	**¥298.26**	**-1%**
United States (USD)	¥41.78	¥226.30	¥214.83	¥26.22	¥37.83	**¥546.96**	¥2.50	**¥574.27**	**+90%**

Per-region should-cost using regional labour, energy and rate benchmarks. Tooling held fixed. Ex-Works — excludes import duty + international freight. Indicative — confirm with RFQ.

§10 — Embodied Carbon

19.74 kgCO2e · 6.24 /kg · cradle-to-gate

Material	**7.18 kgCO2e**	Material factor	**2.10 kg/kg (Steel)**
Process	**12.05 kgCO2e**	Grid intensity	**0.550 kg/kWh · 21.91 kWh**
Logistics	**0.51 kgCO2e**	Total	**19.74 kgCO2e**

Indicative cradle-to-gate factors — replace with supplier EPDs for formal reporting.

§11 — Cost Intelligence Insights

6 findings · ~17% combined saving

[WARNING] Machining / process cost above benchmark

Finding	Process cost at 48.5% of total is above the benchmark range of 8–16%. Cycle time or machine rate is elevated.
Impact	High · up to 15% potential saving
Benchmark	Process %: yours 48.5% vs industry 8–16%
Action 1	Reduce cycle time via feeds/speeds optimisation and toolpath strategy review
Action 2	Combine multiple operations into one setup to eliminate re-fixture time

[WARNING] Elevated process cost — possible high reject/scrap rate

Finding	Process cost is 48.5% vs benchmark max of 16%. For castings, this may indicate a high reject rate (>5%) inflating effective machine time, poor OEE on the casting cell, or cycle time above industry benchmark for this alloy/weight range.
Impact	Medium · up to 8% potential saving
Action 1	Audit actual first-pass yield (FPY) data from the foundry — target >96% for HPDC Al
Action 2	Root-cause the dominant reject type: porosity, dimensional, surface, or cold shut

[WARNING] Low material utilisation — high scrap rate

Finding	Material utilisation at 65% is below the casting benchmark of 80%. You are paying for 54% more raw material than ends up in the part.
Impact	Medium · up to 3% potential saving
Action 1	Redesign blank/preform geometry to minimise offcuts
Action 2	Optimise nesting/layout for sheet or profile stock

[WARNING] Low casting yield — excess runner/gating material

Finding	Casting yield of 65% means 54% of poured metal is returned as runner/gate scrap. HPDC benchmark is 65–75%; sand/gravity 75–85%. Scrap recovery credits are partial — you are paying for metal that doesn't end up in the part.
Impact	Medium · up to 1% potential saving
Benchmark	Casting Yield: yours 65.0% vs industry 65–85%
Action 1	Work with die designer to optimise runner/gate geometry and reduce pour weight
Action 2	Consider vacuum-assisted HPDC (vacural) to allow thinner gates and reduce gating volume

[INFO] Material cost is below benchmark range

Finding	Material at 12.2% is below the typical 40–65% range. Conversion cost is the dominant driver.

Impact	Low
Benchmark	Material %: yours 12.2% vs industry 40–65%
Action 1	Focus optimisation efforts on process and labour efficiency

[INFO] Confirm post-casting scope — no finishing operations in this cost

Finding	The operation list carries no heat treatment, shot blasting or fettling line. If the casting form's service adders were set they are already in the material bucket; if not, structural castings typically add 8–18% for T5/T6 heat treatment (£1.20–2.80/kg), shot blast (£0.15–0.40/part), impregnation (£0.80–1.80/part) and deburring (£0.10–0.60/part).
Impact	Medium
Action 1	Check the derivation trace: heat-treat/descale/NDT adders appear there when they were costed
Action 2	If genuinely absent, set the casting service adders (heat treat, shot blast, impregnation) and re-cost

§12 — Cost-Reduction Opportunities (ranked by saving)

6 opportunities · combined ¥52.85/part (~18%)

Ranked on money per part against this costing. The combined figure is the root-sum-square of the top three, capped at 40% — overlapping actions do not save twice, so the column is deliberately not summed.

#	Category	Action	Why it is on the list	Save/part	%	Timeframe	Risk
1	Process & Automation	**Multi-Cavity / Multi-Up Tooling**	Every cycle currently makes one part. A 2-cavity (or family) tool halves the machine and labour minutes per part against roughly 60–80% more tooling NRE.	¥36.24	12.0 %	Medium Term	Medium
2	Process & Automation	**Attack the Bottleneck: "CNC Machining — 29 features (2×Ø8.0×18, 2×Ø8.0×23, 2×Ø10.5×13, 2×Ø14.0×10, 2×Ø22.0×40, 1×Ø24.0×16, 1×Ø43.0×8, 1×Ø50.0×13, 2×Ø50.0×32, 2× face 2915mm², 2× face 2551mm², 2× face 1986mm², 2× face 1423mm², 1× face 1357mm², 1× face 868mm², 2× face 698mm², 2× face 695mm²) [geometry-measured]"**	One operation — CNC Machining — 29 features (2×Ø8.0×18, 2×Ø8.0×23, 2×Ø10.5×13, 2×Ø14.0×10, 2×Ø22.0×40, 1×Ø24.0×16, 1×Ø43.0×8, 1×Ø50.0×13, 2×Ø50.0×32, 2× face 2915mm², 2× face 2551mm², 2× face 1986mm², 2× face 1423mm², 1× face 1357mm², 1× face 868mm², 2× face 698mm², 2× face 695mm²) [geometry-measured] — carries 74% of the conversion cost. Improving anything else first is working on the wrong problem.	¥24.16	8.0%	Medium Term	Medium
3	Process & Automation	**Multi-Machine Manning**	Operations are manned at up to 1.0 operators per machine while the machine paces the cycle. One operator tending two machines is the textbook correction.	¥11.36	3.8%	Quick Win	Low
4	Material	**Improve Material Utilisation via Near-Net-Shape**	Current utilisation at 65%. Near-net-shape process or improved nesting targets 80–90%.	¥30.20	10.0 %	Medium Term	Medium
5	Material	**Consumables Rationalisation (Cores, Patterns, Shell)**	Per-part consumables are 37% of the material line. Core count, pattern life and shell recipe are all design-controllable.	¥18.12	6.0%	Medium Term	Medium
6	Sustainability & Energy	**Energy Productivity Programme**	This process is energy-intensive (melt, cure, oven or heat-treat content). Energy sits inside the machine rate and overhead — and it is the fastest-moving input cost of the decade.	¥9.06	3.0%	Medium Term	Low

§13 — Opportunity by Category

3 categories with an actionable lever

Category	Opportunities	Best single action	Best save/part	Category total (indicative)
Process & Automation	3	Multi-Cavity / Multi-Up Tooling	**¥36.24**	¥71.76
Material	2	Improve Material Utilisation via Near-Net-Shape	**¥30.20**	¥48.32
Sustainability & Energy	1	Energy Productivity Programme	**¥9.06**	¥9.06

§14 — Inputs to Confirm

No saving is claimed against these

Item	What the costing assumed	How to close it
Unusually low material content — verify alloy grade	Material at 12.2% of part cost is below the 35% floor for castings. Verify alloy grade and net weight inputs.	Validate material weight and alloy price inputs against purchase orders

§15 — Implementation Roadmap

Quick Wins — Implement Immediately

- Multi-Machine Manning